Supplemental Material

Crowdsourcing Cognitive Maps for Generative AI

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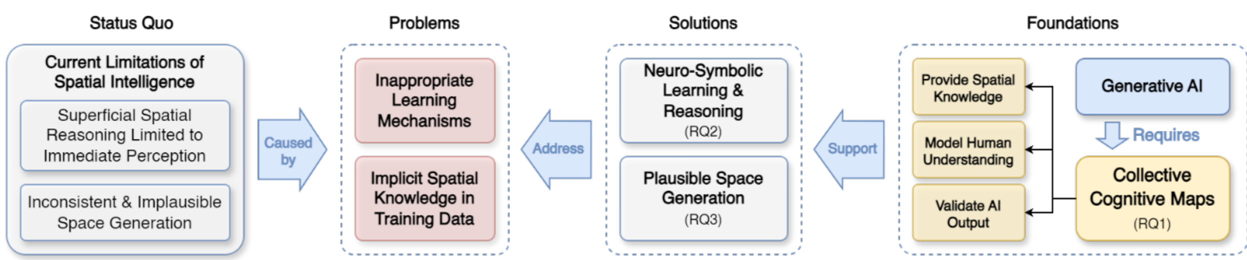
Sihang Qiu, Xiaolong Zheng*, Dingqi Yang, Zhengqiu Zhu, Yong Zhao, Xiaohui Wang, Qingpeng Zhang, Daniel Dajun Zeng



Graphical Abstract: An outline illustrating the current limitations of the AI's spatial intelligence, explaining key problems causing these limitations, proposing potential solutions for addressing the problems, and highlighting the foundational role that cognitive maps could play in providing promising solutions together with generative AI. Corresponding research questions (RQ1, RQ2, and RQ3) are detailed in the main text.

1 Cognitive Map Language (CML)

1.1 CML Elements

We propose an XML-based Language - Cognitive Map Language (CML), for describing Collective Cognitive Maps (CCM). The language employs the following elements:

- **Nodes** represent entities within the cognitive map, including physical objects, spatial points, events, decisions, or abstract concepts. For instance, in a cognitive map of a campus, a node could represent the building of a college, or in a map of an emergency management process, a node could represent the decision for evacuation.
- **Relations** capture the spatial and logical connections between elements. Nodes and relations are two common types of elements widely used in existing topological maps, scene graphs, and other knowledge graphs.

- **Edges** define linear features and boundaries within the cognitive map (also widely used in other map structures, referred to as "ways"). These edges can describe segments and polygons, and they can be either physical or conceptual.
- **Districts** represent functional areas that contain various spatial elements. Importantly, a district can recursively contain a complete CCM structure of its own, thereby enabling the hierarchical nature of the CCM.
- **References** denote external cues that aid in constructing cognitive maps and activities such as navigation or decision-making (landmarks in Lynch's theory [1]). These references can include physical landmarks or other salient signs, and they uniquely distinguish cognitive maps from other map types.
- **Paths** are core elements that distinguish the CCM from similar concepts. Paths are described by a sequence of "element-action" steps. Paths form the skeleton (main functions) of a cognitive map, while the other five elements serve as its flesh and blood. These paths can represent activities, workflows, tasks, or even abstract decision-making processes.

1.2 CML Schema Definition

We propose an XML Scheme Definition (XSD) of CML, to strictly define how to describe spaces using CML. This XSD allows LLMs to organize spatial information in a standard format, and more importantly, inspire LLMs to provide contextual knowledge they would not typically produce on their own.

```xml
<?xml version="1.0" encoding="UTF-8"?>
<xs:schema xmlns:xs="http://www.w3.org/2001/XMLSchema">

    <!--
    ========================================================
    REUSABLE BASE TYPES
    ========================================================
    A base type for structural elements (Node, District, Reference) that share
    the common attributes: id, name, quantity, and size.
    -->
    <xs:complexType name="MapElementBaseType">
        <xs:attribute name="id" type="xs:string" use="required"/>
        <xs:attribute name="name" type="xs:string" use="required"/>
        <xs:attribute name="quantity" type="xs:string" use="optional"/>
        <xs:attribute name="size" type="xs:string" use="optional"/>
        <xs:anyAttribute processContents="lax" namespace="##any"/>
    </xs:complexType>

    <!--
    ========================================================
```

```
    1 & 2. NODE and REFERENCE (Landmark) Elements
    ========================================================
    Both are defined using the common MapElementBaseType.
    -->
    <xs:element name="Node" type="MapElementBaseType"/>
    <xs:element name="Reference" type="MapElementBaseType"/>

    <!--
    ========================================================
    3. DISTRICT
    ========================================================
    A District is an extension of the base type and includes a sequence of
    ChildElement(s) which can be other Nodes, References, or Districts (recursive).
    -->
    <xs:complexType name="DistrictType">
        <xs:complexContent>
            <xs:extension base="MapElementBaseType">
                <xs:sequence>
                    <xs:choice minOccurs="0" maxOccurs="unbounded">
                        <xs:element ref="Node"/>
                        <xs:element ref="District"/>
                        <xs:element ref="Reference"/>
                    </xs:choice>
                </xs:sequence>
                <xs:attribute name="districtType" type="xs:string" use="required" />
                <xs:anyAttribute processContents="lax" namespace="##any"/>
            </xs:extension>
        </xs:complexContent>
    </xs:complexType>
    <xs:element name="District" type="DistrictType"/>


    <!--
    ========================================================
    4. RELATION (RDF-like)
    ========================================================
    Defines a relation with attributes for type, subject ID, and object ID,
    making it easily transformable into a Subject-Predicate-Object (SPO) triple.
    -->
    <xs:complexType name="RelationType">
        <xs:attribute name="id" type="xs:string" use="required"/>
        <xs:attribute name="relationType" type="xs:string" use="required"/>
        <xs:attribute name="subjectId" type="xs:string" use="required"/>
        <xs:attribute name="objectId" type="xs:string" use="required"/>
        <xs:anyAttribute processContents="lax" namespace="##any"/>
    </xs:complexType>
    <xs:element name="Relation" type="RelationType" />


    <!--
    ========================================================
    5. EDGE
    ========================================================
    Defines an Edge that separates two logical groups of elements.
    An edge can be passable (door, bridge, tunnel, etc.) or not (wall, river,
mountain, etc.).
    -->
    <xs:complexType name="ElementGroupType">
        <xs:sequence>
            <!-- A group consists of one or more references to element IDs -->
            <xs:element name="ElementId" type="xs:string" maxOccurs="unbounded"/>
        </xs:sequence>
    </xs:complexType>
```

```
    <xs:complexType name="EdgeType">
        <xs:sequence>
            <xs:element name="GroupA" type="ElementGroupType"/>
            <xs:element name="GroupB" type="ElementGroupType"/>
        </xs:sequence>
        <xs:attribute name="id" type="xs:string" use="required"/>
        <xs:attribute name="name" type="xs:string" use="required"/>
        <xs:attribute name="edgeType" type="xs:string" use="required"/>
        <xs:attribute name="size" type="xs:string" use="optional"/>
    </xs:complexType>
    <xs:element name="Edge" type="EdgeType"/>


    <!--
    ========================================================
    6. PATH (Sequence of Activities)
    ========================================================
    A Path defines an ordered sequence of Steps, where each step specifies an
element
    and an activity to be performed there.
    -->
    <xs:complexType name="PathStepType">
        <xs:attribute name="elementId" type="xs:string" use="required"/>
        <xs:attribute name="activity" type="xs:string" use="required"/>
    </xs:complexType>

    <xs:complexType name="PathType">
        <xs:sequence>
            <xs:element name="Step" type="PathStepType" maxOccurs="unbounded"/>
        </xs:sequence>
        <xs:attribute name="id" type="xs:string" use="required"/>
        <xs:attribute name="name" type="xs:string" use="required"/>
        <xs:attribute name="pathType" type="xs:string" use="required"/>
        <xs:anyAttribute processContents="lax" namespace="##any"/>
    </xs:complexType>
    <xs:element name="Path" type="PathType"/>


    <!--
    ========================================================
    ROOT ELEMENT: CognitiveMap
    ========================================================
    The main container for all the map's structural and relational elements.
    -->
    <xs:element name="CognitiveMap">
        <xs:complexType>
            <xs:sequence>
                <xs:choice minOccurs="0" maxOccurs="unbounded">
                    <xs:element ref="Node"/>
                    <xs:element ref="District"/>
                    <xs:element ref="Relation"/>
                    <xs:element ref="Reference"/>
                    <xs:element ref="Edge"/>
                    <xs:element ref="Path"/>
                </xs:choice>
            </xs:sequence>
        </xs:complexType>
    </xs:element>

</xs:schema>
```

1.3 Tools for CML

We provide multiple tools for parsing, visualizing, and using CCMs based on CML (https://github.com/qiusihang/cognitive-map). The paring tool (src/cognitive_map.py) uses the default XML library provided by Python3 to read and load the CCM.

The map generation & visualization tools (src/map_generation.py & src/map_visualizer.py, run map_generation.py only) employ a two-phase force-directed layout algorithm to generate spatially consistent random maps based on the given cognitive map. In Phase 1, CCM element centers are positioned through iterative simulation using directional attraction forces (based on semantic relations like "north" or "near") and global centering forces, establishing a stable topological configuration. Phase 2 performs gradual object growth from a point to full scale while resolving collisions through hard boundary correction and soft repulsion between unrelated objects, maintaining aspect ratios and relational constraints. The process runs recursively for hierarchical structures (i.e., for District elements), ensuring both global coherence and local detail preservation.

The questions for crowdsourcing microtasks and knowledge for enhancing AI are generated using a generation tool (src/question_generation.py). It uses Table 3 of the main-text to create conversational questions and natural-language knowledge for crowdsourcing and CCM application respectively.

2 Preliminary Experiments

We carried out 4 studies as preliminary experiments to show the feasibility of crowdsourcing cognitive maps. Study I and Study II aims at the crowdsourcing process (RQ1 in the main-text), while Study III and Study IV, respectively, focus on using CCM to improve spatial thinking abilities of AI (RQ2) and space generation abilities of AI (RQ3). Studies I, II, and IV involve human participants from the platfrom Prolific. These studies were approved by our IRB. All participants provided informed consent, and we did not collect personal data.

2.1 Study I: Feasibility of Human-AI Collaborative Crowdsourcing

The objective of pilot study I is to test whether a hybrid human-AI workflow can effectively build and refine a CCM. We let an LLM (Qwen3-235B) generate an initial CCM for a "coffee shop" with the XSD included in the model input (the XSD inclusion not only help in formalizing information in the standard format, but also inspire LLMs to provide contextual knowledge they would not typically produce on their own). This initial map (Initially 7 nodes, 3 relations, 1 edges, 4 districts, 2 paths – 5 and 3 steps for each path respectively) was broken down into 21 microtasks (using Table 3). We recruited 42 participants asking them to complete 5 microtasks each, resulting in 210 microtask answers -- with each microtask answered by 10 workers. A conversational user interface is deployed to execute the task. Conflicts (the threshold is that at least 3 participants disagree) were resolved by prompting the LLM to update the CCM based on the human feedback. The prompt for the LLM is:

```
You are tasked with improving an XML file that was created according to
the provided XSD schema.
Your focus is on the following element IDs: [IDS]

Please follow this process:
- Analyze the feedback from people: [FEEDBACK]
- Identify common themes and consensus in the feedback (at least
proposed by two people simultaneously)
- Check whether the content of the feedback has similar or conflicting
meanings with existing elements. If so, do not consider this feedback.
- Determine the required actions: decide whether to add, update, or
delete the specified elements based on the feedback analysis
- Implement changes: modify the target elements and update any other
elements that reference them
- Ensure compliance: all modifications must adhere to the XSD schema

Materials provided:
Complete XML file:
<XML-CONTENT>

XSD schema:
<XSD-CONTENT>

Return the improved XML file with your changes implemented.
```

Results:

The CCM was enriched through crowdsourcing, growing from 17 to 28 elements (11 nodes, 8 relations, 1 edge, 6 districts, and 2 paths – 4 and 5 steps for each path respectively).

- **Agreement:** We observed a percent agreement of 0.58. The Fleiss' Kappa is 0.092, which is low due to high agreement on the option of "no change needed" for many items, creating a skewed distribution — a common situation in refinement tasks that limits this statistic's utility. While this pilot used single-pass crowdsourcing, a more robust design would iteratively crowdsource and refine the CCM until it achieves an acceptable level of agreement (e.g., percent agreement > 80%).
- **Cost/Time:** The target hourly wage was set at £20/h, though the actual wage paid was £23.86/h. Given that the study (comprising three microtasks) was completed in 151 seconds on average (50 seconds per microtask), the total cost for building a CCM of a coffee shop (inclusive of platform fees) was £56. Such a cost is likely acceptable for industrial applications that require crowdsourcing specific scenarios for AI agents. However, for large-scale campaigns, we recommend establishing a CCM ecosystem relying on voluntary contributions, as detailed in our manuscript.
- **Effectiveness:** This crowdsourced CCM was successfully used in our AI tasking experiments (addressing RQ2/RQ3), demonstrating its practical effectiveness. We show this crowdsourced CCM can significantly improve the task completion efficiency in a simulated scenario. We also show significant improvement in the quality of space generation, in comparison with an unconstrainted baseline.

2.2 Study II: Capturing Human Spatial Understanding

The objective of pilot study II is to validate that CCMs can capture human spatial understanding and quantify differences/biases between groups.

According to UN Geoscheme of "Western European", we focus on people's spatial understanding of the eight countries: Austria, Belgium, France, Germany, Liechtenstein, Luxembourg, Monaco, Netherlands, and Switzerland. This study investigates the perceptions of two major English-speaking demographic groups (from the UK and the US) regarding this geographical area. We create 36 microtasks for the "Western Europe" CCM – 9 about basic attribute and $\binom{9}{2} = 36$

relations. We recruited two distinct groups from Prolific: 27 native English speakers from the UK and 27 from the US. Each participant was asked to complete 5 microtasks. Therefore, each microtask receives 27*5/45=3 answers. The average completion time is 124.5s and the average hourly range is £28.92/h.

Results: We found clear quantitative and visual differences between the UK and US group's spatial understanding of "Western Europe". The UK group showed higher internal agreement compared to US group (Percent Agreement: 0.415 vs. 0.296, Fleiss Kappa: 0.307 vs. 0.143), suggesting a more consolidated cultural understanding of the region (that is closer to UK). As can be seen in the visualization of CCMs (Figure S1), it is clear that UK group showed a more precise understanding of Western Europe in terms of the area and spatial layout, in comparison with US group. Therefore, we envision that CCM can be a very useful tool for capturing and comparing spatial understandings of specific groups of people. However, we must acknowledge that this is a simple pilot – the sample sizes are still small and other demographic factors such as education were not considered. This study serves as a preliminary subjectivity/bias analysis, demonstrating that CCMs can reflect biases of specific group of different demographic and cultural background. We plan to further develop and refine the CCM as a professional tool to quantitatively measure cognitive differences.

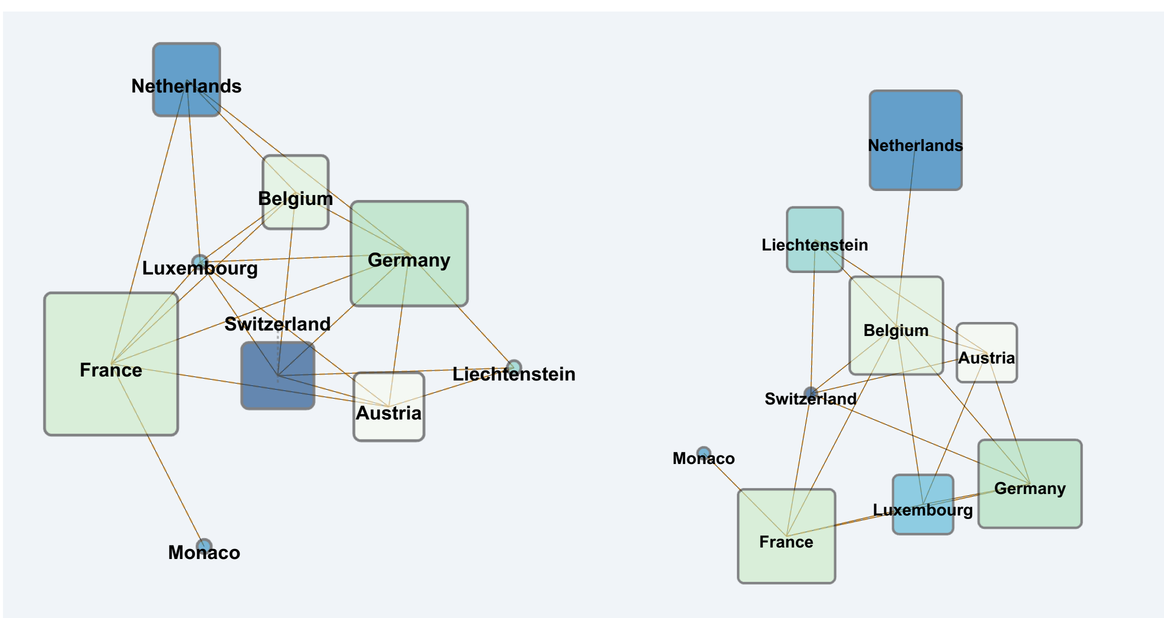


Figure S1 The visualization of "Western Europe" CCMs crowdsourced by UK participants (left, n=27) and US participants (right, n=27).

2.3 Study III: Using CCM Knowledge to Improve AI

The task for AI to do in this experiment is to let an AI agent enter a coffee shop to buy a coffee and take it back in a simulated scenario. The objective is to test an AI agent's ability of spatial reasoning, specifically to test: whether AI can find the correct path to order the coffee, then complete a sequence of activities until it gets the coffee, and finally find its way back. The scenario layout is shown in Figure S2.

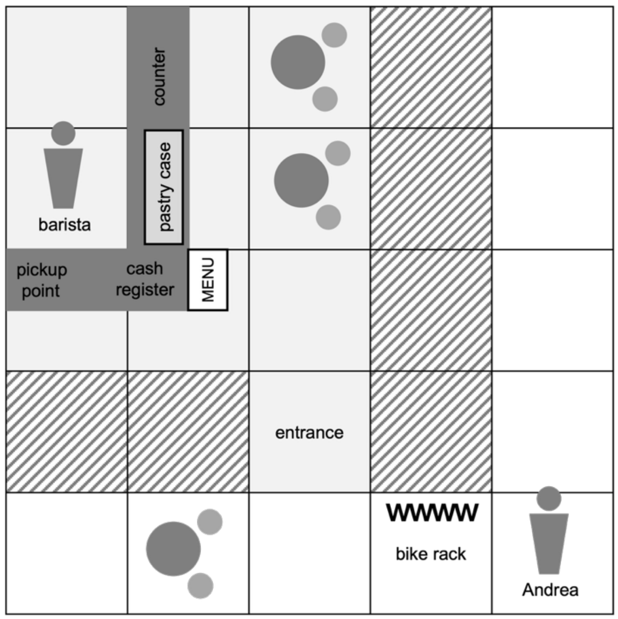


Figure S2 The layout of the coffee shop scene for text-based simulation

To isolate spatial reasoning from visual perception, we used a text-based simulation environment. The agent receives a textual description of its surrounding environment (an interactive demo can be found on our website). This mirrors the core reasoning process of Vision-Language-Action (VLA) models, which ultimately rely on a textual representation for reasoning and decision-making.

Experimental Conditions:

- **Instruction Type:** simple instruction without spatial clues (Mission 1), and instructions with spatial clues (Mission 2). The latter is the typical mission type in the most prevalent AI studies.
- **Access to CCM:** we compared agent performance with and without access to a CCM describing coffee shops.

We implemented a CCM-augmented pipeline as follows:

1. Knowledge Generation: The coffee shop CCM was converted into a structured text description (see Table 3 in the manuscript).
2. Retrieval-Augmented Generation (RAG): The agent's mission triggered a retrieval step from a vector database of the CCM knowledge. The LLM was then prompted with the relevant contextual knowledge provided by the CCM.

Evaluation Metrics:

- **Success Rate:** The percentage of runs in which the agent successfully takes the coffee back. A run is considered to be failed if the agent does not succeed within 30 steps.
- **Steps Taken:** The average number of steps required for successfully taking the coffee back. A lower value indicates higher efficiency.

After 50 repeated runs for each of the 2x2 conditions, we acquired the results as shown in Table S1.

Table S1 Results of text-based simulation to test spatial-reasoning abilities of AI

		Mission 1 (without spatial clues) *"Please get me an Americano. Buy it from the coffee shop and give it to me."*	**Mission 2 (with spatial clues)** *"Please get me an Americano. Here's what to do: go in through the entrance, check the menu for Americanos, order and pay at the cash register, wait for it to be made, then bring it to me from the pickup point."*
LLM w/ CCM	**Success Rate**	100%	100%
	Steps Taken	15.60 ± 0.61	14.06 ± 0.24
LLM w/o CCM	**Success Rate**	24%	100%
	Steps Taken	18.75 ± 1.36	14.80 ± 0.66

The results reveal a statistically significant difference in the number of steps taken to complete the tasks across all experimental groups (all p-values < 1.8e-9, Mann-Whitney U tests, Bonferroni-corrected). This confirms that LLMs can complete tasks efficiently when provided with explicit spatial clues. Importantly, the introduction of the CCM via RAG significantly improved task efficiency (i.e., reduced the number of steps) for both mission types. This demonstrates that the CCM effectively provides the missing contextual knowledge, enabling more efficient and reliable task completion by generative AI. We acknowledge this study employs a simplified setting. Future work will build upon these promising results by randomizing environmental scenes, introducing a wider variety of complex tasks, and incorporating more nuanced metrics for evaluation, such as human-like spatial perception.

Key Findings of LLMs without the CCM:

- **Rely on Explicit Spatial Clues:** The agent succeeded in a controlled baseline task where instructions included explicit spatial clues (e.g., "pay at the cash register"), confirming that the failure mode is not navigation but reasoning about implicit procedures.
- **Fail to Infer Multi-Step Activities:** Without the CCM, the agent often failed to infer the unstated steps of buying a coffee. While it could perform the obvious action ("order coffee"), it frequently omitted critical steps like paying at the counter, waiting for the coffee to be made, and picking it up from the service area. It would often assume the task was complete after ordering and attempt to leave.
- **Fail to Sequence Actions Correctly:** Even when some steps were performed, the agent without the CCM demonstrated poor action sequencing, such as trying to pick up the coffee before it was paid, or repeating invalid actions. This indicates a lack of understanding of the functional and causal relationships between spatial elements and activities.

This study provides concrete evidence that current LLM-based agents struggle with the implicit spatial and functional knowledge that humans use effortlessly, and it demonstrates how a CCM can directly supply this missing context to overcome the failure.

2.4 Study IV: Using CCM Knowledge to Improve Space Generation

We evaluated the use of the crowdsourced CCM as a knowledge source for generative models to produce space of coffee shops. We generated random images of the space of the coffee shop using Gemini 2.5 Flash Image, either with or without CCM guidance. CCM guidance is presented in LLM prompts as shown below.

Prompt (w/o CCM):

```
Create a photorealistic, top-down view of a coffee shop, with no text,
no label, and no people.
```

Prompt (w/ CCM):

```
Create a photorealistic, top-down view of a coffee shop, with no text,
no label, and no people.
Here is some contextual information you should consider during the
image generation process: [spatial knowledge exported from the CCM]
```

We perform a human evaluation study on Prolific, participants (n=50 per metric) compared pairs of generated spaces (w/ CCM vs. w/o CCM baseline, 5 each, as shown in Figure S3) on three key metrics: contextual richness, spatial coherence, and functional plausibility. The average hourly wage is £28.57, and average task execution time (for comparing one pair) is 63s. The task interface for comparing scenes is shown in Figure S4.

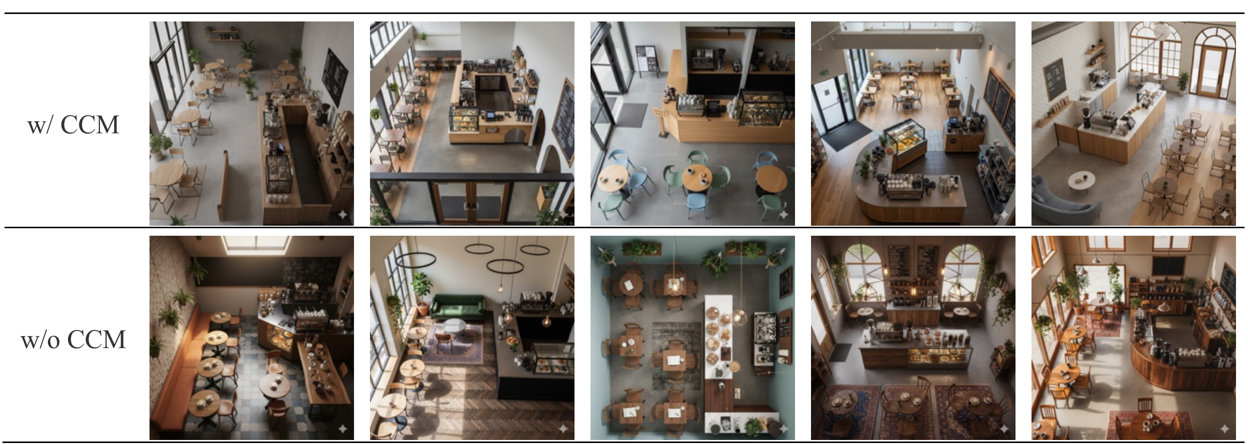


Figure S4 Comparisons of coffee shops generated by generative AI with or without CCM. They all look like coffee shops. However, after human evaluation (n=50), the CCM-guided data generation shows a statistical win on spatial coherence and functional plausibility.

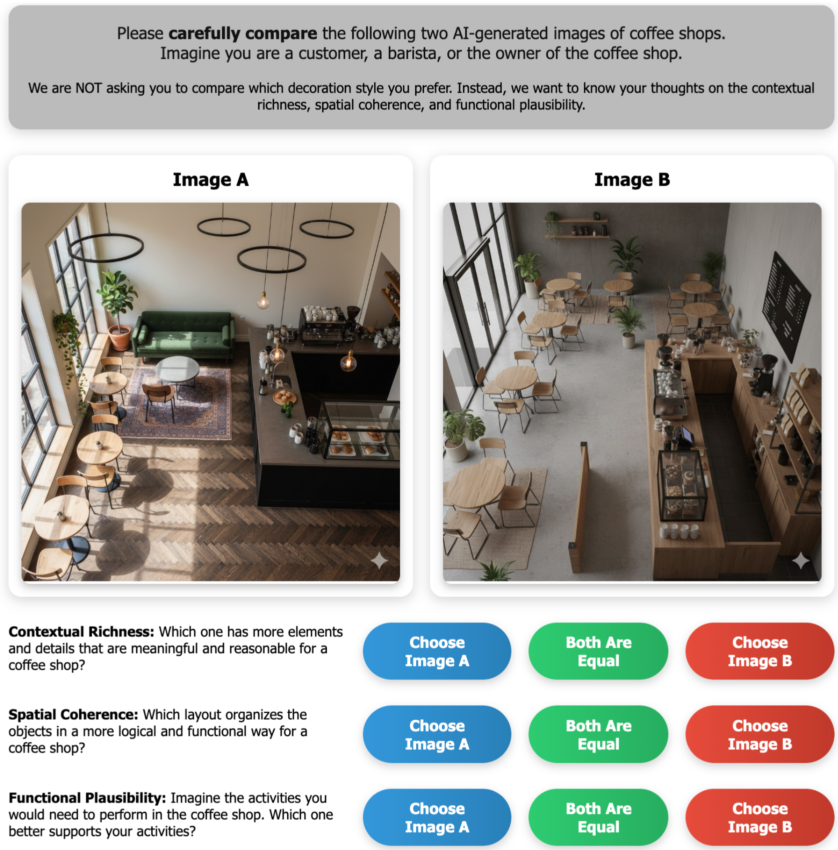


Figure S4 The task interface for comparing AI-generated spaces of coffee shops.

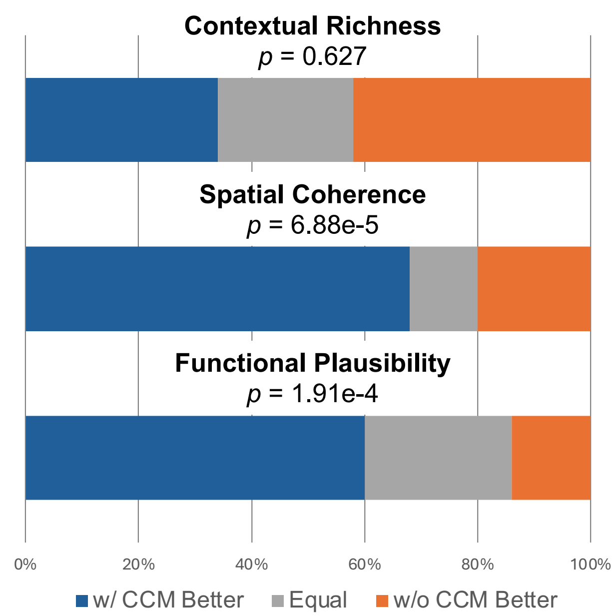


Figure S5 Human-evaluated results of space generation.

Results: The CCM-guided approach demonstrated a clear and statistically significant win on major spatial metrics (as shown in Figure S5).

- **Contextual Richness:** No significant difference was found, indicating that current image generator can already generate rich contextual details in a "coffee shop" scene without the guidance of CCMs.
- **Spatial Coherence:** CCM-guided spaces were strongly preferred by participants (Binomial test, $p=6.88e-5$), indicating their layouts organize the objects in a more logical and functional way for a coffee shop.
- **Functional Plausibility:** CCM-guided spaces were strongly preferred (Binomial test, $p=1.91e-4$), indicating they better supports activities need to perform in a coffee shop.

These results provide evidence that CCMs can effectively guide generative models to produce more spatially coherent and functionally plausible environments. The current experiment uses a single-pass generation process where the CCM is used to structure the initial prompt. Looking forward, we envision a more powerful, iterative generation and verification loop.

3 Applications

This enhanced spatial understanding will open up a wide array of applications, including training autonomous systems, revolutionizing design industries, and providing new avenues for simulation in scientific experiments.

3.1 Training Autonomous Systems

In the realm of autonomous systems, cognitive maps will improve operations, question answering, and navigation of AI-embodied robots/agents [2], [3]. Generating effective 3D training environments, particularly for complex tasks, presents a significant challenge for current generative AI models. We argue that CCMs can be used to generate consistent and plausible virtual training spaces, empowering embodied AI agents to learn complex tasks, navigate complex environments, understand spatial commands, and provide contextually relevant information about their surroundings.

3.2 Revolutionizing Design Industries

In spanning 3D design industries like manufacturing, architecture, film, and gaming, generative AI can already facilitate intuitive creation and manipulation of virtual spaces [4]. With cognitive maps equipped by AI, designers can create more dynamic and believable 3D worlds, where characters and environments behave more realistically. Similar to what it can do in agent training, it can also create more immersive and engaging virtual experiences for human users by simulating realistic environments and interactions. With generative AI enhanced by cognitive maps, it will enable designers with limited technical expertise to create complex environments, significantly lowering the barrier for advanced design.

3.3 Reliable Simulation for Scientific Experiments and Decision-Making

This technology will offer exciting possibilities for scientific research [5]. Particularly, we emphasize its importance in social simulation for better understanding the society where we are living in. By integrating cognitive maps with generative AI, researchers and decision makers can create simulations of complex social environments inhabited by AI agents that behave more like humans. This creates a powerful new testbed for social experiments, allowing researchers and decision makers to explore hypotheses and strategies, and gain reliable predictions about their potential consequences. This new approach can provide traditional social experiments with valuable supplementary computational evidence and can significantly reduce experimental costs.

4 CCM Ecosystem

The ultimate goal is to build an ecosystem for crowdsourcing cognitive maps, as shown in Figure S6. It will benefit various communities (academic, education, creator, developer, etc.), by providing a basis for generating various spaces or training AI spatial thinking abilities. Members in these communities actively provide crowdsourcing contributions back to the ecosystem in return, forming a self-sustaining model similar to Wikipedia. Sustaining such a flourishing ecosystem needs both researchers and practitioners to devote themselves to tackling problems such as launching the crowdsourcing campaign, ensuring data quality and consistency, maintaining participation, protecting user privacy, and managing online systems (similar to Git) for storing cognitive maps, which must rigorously track provenance, cultural stratification, versioning, branching, and all contribution workflows.

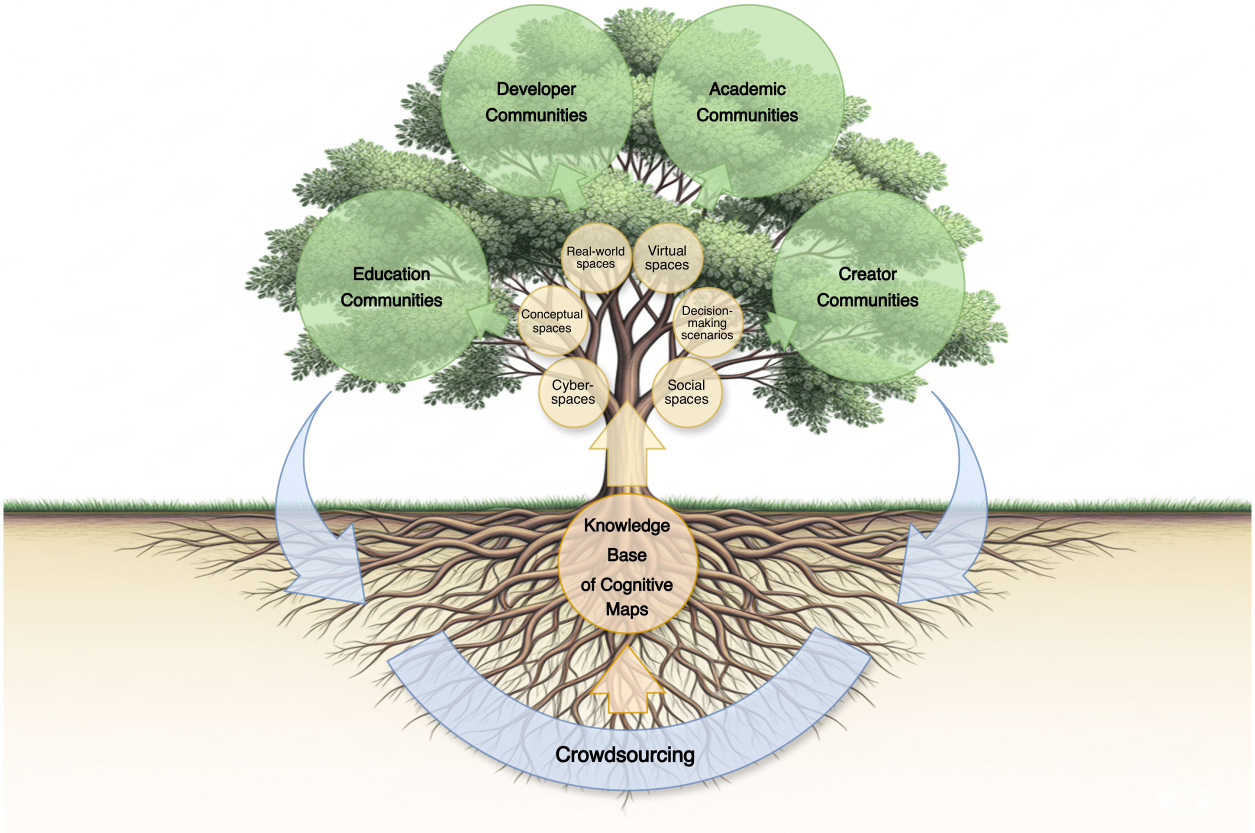


Figure S6 The ecosystem of crowdsourcing cognitive maps. The knowledge base of cognitive maps will serve diverse virtual space generation and spatial cognition training, further benefiting academic communities, education communities, creator communities, developer communities, and more. Active members of these communities will, in return, contribute back to the knowledge base through voluntary crowdsourcing, forming a flourishing ecosystem.

5 Data & Code Availability

All the data and code of this work are available at: https://github.com/qiusihang/cognitive-map.

References

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